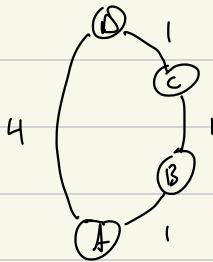


Q9

This is false. Here is a counter-example:

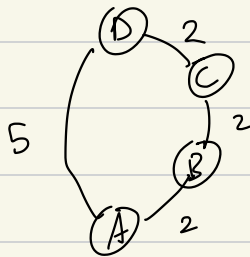
Original graph:



Shortest path: ABCD

weight: 3

New graph:



Shortest path: AD

weight: 5